Chapter 26

Class III and V Amalgam Restorations

"Tell me and I forget. Teach me and I remember. Involve me and I learn"

—Benjamin Franklin

This chapter presents information about class III and V amalgam restorations. Class III restorations are indicated for defects located on the proximal surface of anterior teeth that do not affect the incisal edge. Part of the facial or the lingual surfaces may also be involved in class III restorations. Class V restorations are indicated to restore defects on the facial or lingual cervical one-third of any tooth.

The class III amalgam restoration is rarely used. Its use has been supplanted by tooth-coloured restorations (primarily composite), which have become increasingly wear-resistant and colour-stable. Because indications exist for class III amalgam restorations, however, practitioners should be familiar with this restorative technique.

The class V amalgam restoration can be especially technique-sensitive because of location, extent of caries, and limited access and visibility. Cervical caries usually develops because of the chronic presence of acidogenic plaque located in the non-self-cleansing area just beneath the coronal height of contour. Patients with gingival recession are predisposed to cervical caries because dentin is more susceptible to demineralization than enamel. Patients with a reduced salivary flow caused by certain medical conditions (e.g. Sjögren's syndrome), medications or head and neck radiation therapy are also predisposed to cervical caries. These patients usually have less saliva to buffer the acids produced by oral bacteria. Patients with gingival recession that has exposed the root surface have a predisposition to *root caries* because dentin is more susceptible to demineralization than enamel. Class V restorations may be used to treat both cervical and root caries lesions.

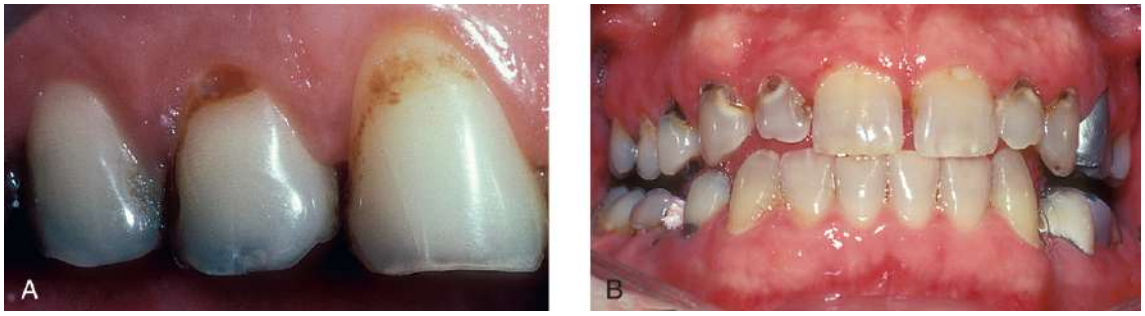
Incipient, smooth-surface enamel caries appears as a chalky white line just occlusal or incisal to the crest of the marginal gingiva (usually on the facial surface) (Online Fig. 26.1). These areas often are overlooked in the oral examination, unless teeth are free of debris, isolated with cotton rolls and dried gently with the air syringe. When incipient cervical caries has not decalcified the enamel sufficiently to result in cavitation (i.e. a break in the continuity of the surface), the lesion may be remineralized by appropriate techniques, including patient motivation towards proper diet and hygiene. Occasionally, an enamel surface that is only slightly cavitated may be treated successfully by smoothing with sandpaper disks, polishing and treating with a fluoride varnish or a dentin adhesive in an attempt to prevent further caries that may require treatment. This prophylactic, preventive treatment cannot be instituted if caries has progressed to decalcify and soften enamel to an appreciable depth. In this instance, a class V tooth preparation and restoration is indicated, particularly if caries has penetrated to the dentinoenamel junction (DEJ) (Online Fig. 26.2A). When numerous cervical lesions are present (Online Fig. 26.2B), a relatively high caries index is obvious. In addition to the restorative treatment, the patient should be instructed and encouraged to implement an aggressive prevention programme to avoid recurrent decay.



Online Figure 26.1 Incipient caries lesions of enamel appear as white spots. The affected surface may be smooth (i.e. non-cavitated). White spots are more visible when dried.
(From Cobourne MT, DiBiase AT: *Handbook of orthodontics*, Edinburgh, 2010, Mosby.)

Pertinent Material Qualities and Properties

Material qualities and properties important for class III and V amalgam restorations are strength, longevity, ease of use and past success. (See Chapter 12 for a discussion of the pertinent material qualities and properties of amalgam.)



Online Figure 26.2 Cervical caries. (A) Cavitation involving enamel and dentin in several teeth. (B) Relatively high caries index is obvious when numerous cervical lesions are present.
(From Perry DA, Beemsterboer PL: *Periodontology for the dental hygienist*, ed 3, St. Louis, MO, 2007, Saunders.)

Indications

Few indications exist for a class III amalgam restoration. It is generally reserved for the distal surface of maxillary and mandibular canines if (i) the preparation is extensive with only minimal facial involvement, (ii) the gingival margin primarily involves cementum or (iii) moisture control is difficult. For esthetic reasons, amalgam rarely is indicated for the proximal surfaces of incisors and the mesial surface of canines.

Class V amalgam restorations may be used anywhere in the mouth. As with class III amalgam restorations, they are generally reserved for non-esthetic areas, for areas where access and visibility are limited and where moisture control is difficult and for areas that are significantly deep gingivally. Because of limited access and visibility, many class V restorations are difficult and present special problems during the preparation and restorative procedures.

One measure of clinical success of cervical amalgam restorations is the length of time the restoration serves without failing (Online Fig. 26.3). Properly placed class V amalgams have the potential to be clinically acceptable for many years. Some cervical amalgam restorations show evidence of failure, however, even after a short period. Inattention to tooth preparation principles, improper manipulation of the restorative material and moisture contamination contribute to early failure. Extended service depends on the operator's care in following accepted treatment techniques and proper care by the patient.

Amalgam may be used on partial denture abutment teeth because amalgam resists wear as clasps move over the restoration. Contours prepared in the restoration to retentive areas for the clasp tips may be achieved relatively easily and maintained when an amalgam restoration is used. Occasionally, amalgam is preferred when the caries lesion extends gingivally enough that a mucoperiosteal flap must be reflected for adequate access and visibility (Online Fig. 26.4). Proper surgical procedures must be followed, including sterile technique, careful soft tissue management and complete debridement of the surgical and operative site before closure.

Contraindications

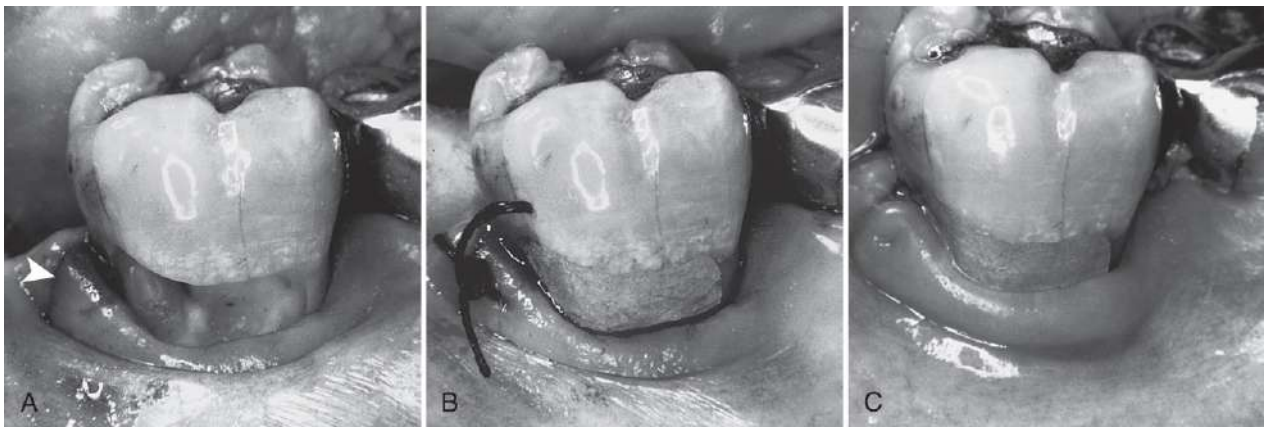
Class III and V amalgam restorations usually are contraindicated in esthetically important areas because many patients object to metal restorations that are visible (Online Fig. 26.5). Generally, class V amalgams placed on the facial surface of mandibular canines, premolars and molars are not readily visible. Amalgams placed on maxillary premolars and first molars may be visible. The patient's esthetic demands should be considered when planning treatment.

Advantages

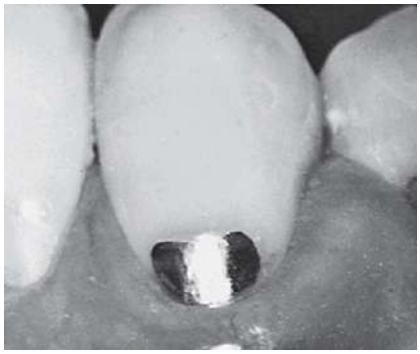
Amalgam restorations are stronger than other class III and V direct restorations. In addition, they are generally easier to place and may be less expensive to the patient. Because of its metallic colour, amalgam is easily distinguished from



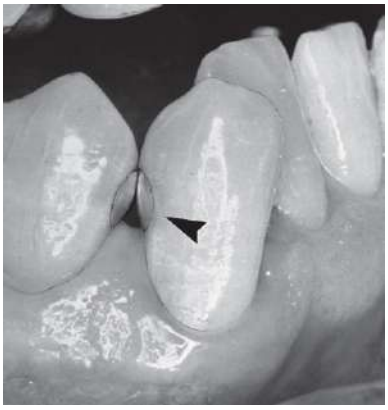
Online Figure 26.3 (A) Six-year-old cervical amalgam restorations. (B) After 16 years, some abrasion and erosion are evident at the gingival margin of the lateral incisor and canine restorations. (C) Twenty-year-old cervical amalgam restorations.



Online Figure 26.4 Surgical access. (A) Class V preparation requiring mucoperiosteal flap reflection with a releasing incision *(arrowhead)*. (B) Completed restoration with suture in place. (C) Suture removed 1 week after the procedure.



Online Figure 26.5 Patients may object to metal restorations that are visible during conversation.



Online Figure 26.6 Restoration for class III tooth preparation using facial approach on mandibular canine. Restoration is 5 years old. (Courtesy of Dr. C. L. Sockwell.)

the surrounding tooth structure. Amalgam restorations are usually easier to finish and polish without damage to the adjacent surfaces.

Disadvantages

The primary disadvantage of class III and V amalgam restorations is that they are metallic and unesthetic. In addition, the preparation for an amalgam restoration typically requires 90-degree cavosurface margins and specific axial depths that allow incorporation of secondary retentive features. These features result in a less conservative preparation than that required for most esthetic restorative materials.

Clinical Technique for Class III Amalgam Restorations

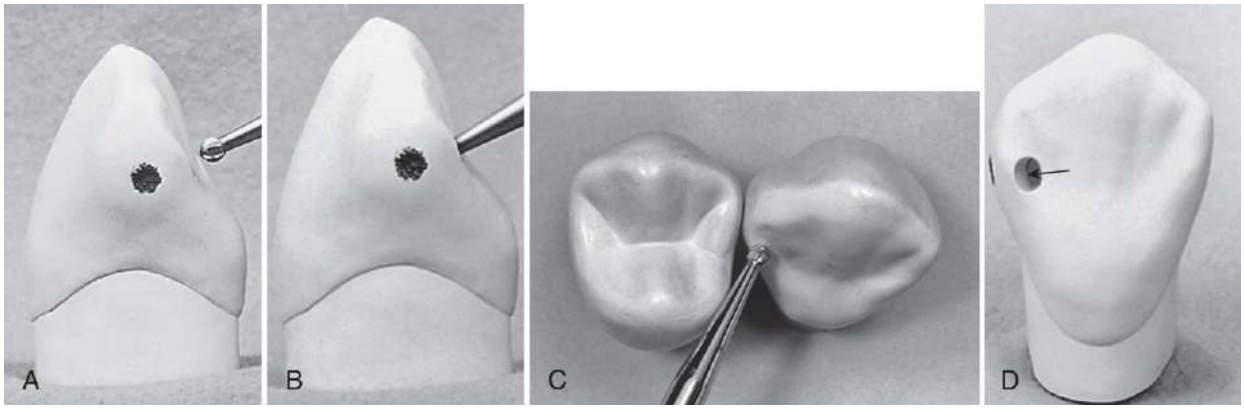
Initial Procedures

After appropriate review of the patient records (including medical history), treatment plan and radiographs, the gingival extension of the preparation should be anticipated. Anaesthesia is usually necessary when a vital tooth is to be restored. Pre-wedging in the gingival embrasure of the proximal site to be restored provides (i) better protection of soft tissue and the rubber dam, (ii) better access because of the slight separation of teeth and (iii) better re-establishment of the proximal contact. The use of a rubber dam is generally recommended; however, cotton roll isolation is acceptable if moisture can be adequately controlled.

Tooth Preparation

A lingual access preparation on the distal surface of the maxillary canine is described here because the use of amalgam in that location is more likely. For esthetic reasons, use of amalgam is best suited for caries that can be accessed from the lingual rather from the facial. A facial approach for a mandibular canine may be indicated, however, if the lesion is more facial than lingual. The mandibular restoration is often not visible at conversational distance (Online Fig. 26.6).

The outline form of the class III amalgam preparation may include only the proximal surface. A lingual dovetail may be indicated if one existed previously or if additional retention is needed for a larger restoration.

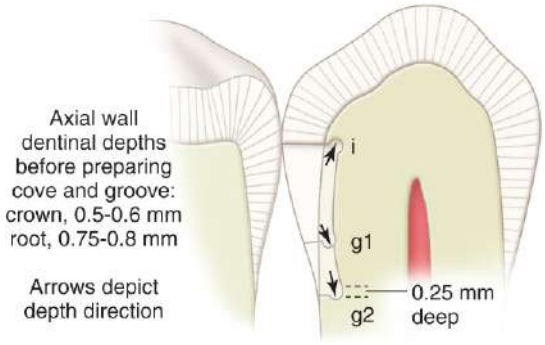


Online Figure 26.7 Entry for class III tooth preparation on maxillary canine. (A) Bur position is perpendicular to the enamel surface at the point of entry. (B) Initial penetration through enamel is directed towards cavitated, caries lesion. (C) Initial entry should isolate the proximal enamel while preserving as much of the marginal ridge as possible. (D) Initial cutting reveals the dentinoenamel junction (DEJ) *(arrow)*.

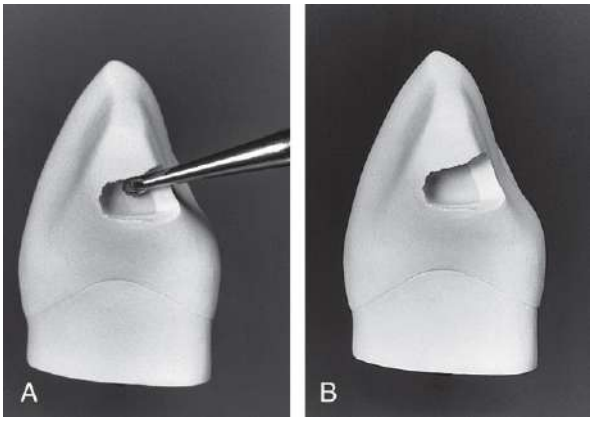
Initial Tooth Preparation

Bur size selection depends on the anticipated size of the lesion. Bur options may include a No. 2 (or smaller) round bur or No. 330 bur. The bur is positioned so that the entry cut penetrates into the caries lesion, which is usually apical to (and slightly into) the contact area. Ideally, the bur is positioned so that its long axis is perpendicular to the lingual surface of the tooth, but directed at a mesial angle as close to the adjacent tooth as possible. (The bur position may be described as perpendicular to the distolingual line angle of the tooth.) This position conserves the marginal ridge enamel (Online Fig. 26.7A–C). Penetration through enamel positions the bur so that additional cutting isolates the proximal enamel affected by caries and removes some or all of the infected dentin. In addition, penetration should be at a limited initial axial depth (i.e. 0.5–0.6 mm) inside the DEJ (Online Fig. 26.7C and D) or at a 0.75–0.8-mm axial depth when the gingival margin is on the root surface (in cementum) (Online Fig. 26.8). This 0.75-mm axial depth on the root surface allows a 0.25-mm distance (the diameter of the No. ¼ bur is 0.5 mm) between the retention groove (which is placed later) and the gingival cavosurface margin. Infected dentin that is deeper than this limited initial axial depth is removed later during final tooth preparation.

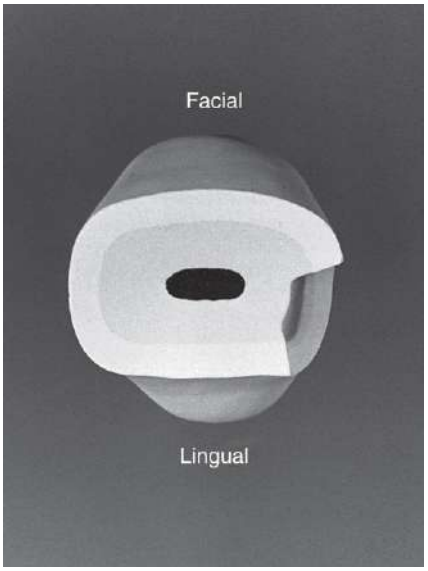
For a small lesion, the facial margin is extended 0.2–0.3 mm into the facial embrasure (if necessary), with a curved outline from the incisal to the gingival margin (resulting in a less visible margin). The lingual outline blends with the incisal and gingival margins in a smooth curve, creating a preparation with little or no lingual wall. The cavosurface angle should be 90 degrees at all margins. The facial, incisal and gingival walls should meet the axial wall at approximately right angles (although the lingual wall meets the axial wall at an obtuse angle or may be continuous with the axial wall) (Online Fig. 26.9). If a large round bur is used, the internal angles are more rounded. The



Online Figure 26.8 Mesiodistal vertical section showing location, depth direction *(arrows)* and direction depth of the retention form in class III tooth preparations of different gingival depths. *i*, Incisal cove; *g1*, gingival groove, enamel margin; *g2*, gingival groove, root surface margin. Distance from outer aspect of *g2* groove to margin is approximately 0.3 mm; bur head diameter is 0.5 mm; direction depth of groove is half this diameter (or approximately 0.3 mm [0.25 mm]).



Online Figure 26.9 Class III tooth preparation on maxillary canine. (A) Round bur shaping the incisal area. The incisal angle remains. (B) Initial shape of the preparation accomplished with a round bur.



Online Figure 26.10 Transverse section of mandibular lateral incisor illustrating that the lingual wall of a class III tooth preparation may meet the axial wall at an obtuse angle and that the axial wall is a uniform depth into dentin and follows the faciolingual contour of the external tooth surface.

axial wall should be uniformly deep into dentin and follow the faciolingual contour of the external tooth surface (Online Fig. 26.10). The initial axial wall depth may be in sound dentin (i.e. shallow lesion), in infected dentin (i.e. moderate to deep lesion) or in existing restorative material, if a restoration is being replaced.

Incisal extension to remove carious tooth structure may eliminate the proximal contact (Online Fig. 26.11). It is important to conserve as much of the distoincisal tooth structure as possible to reduce the risk for subsequent fracture. When possible, it is best to leave the incisal margin in contact with the adjacent tooth.

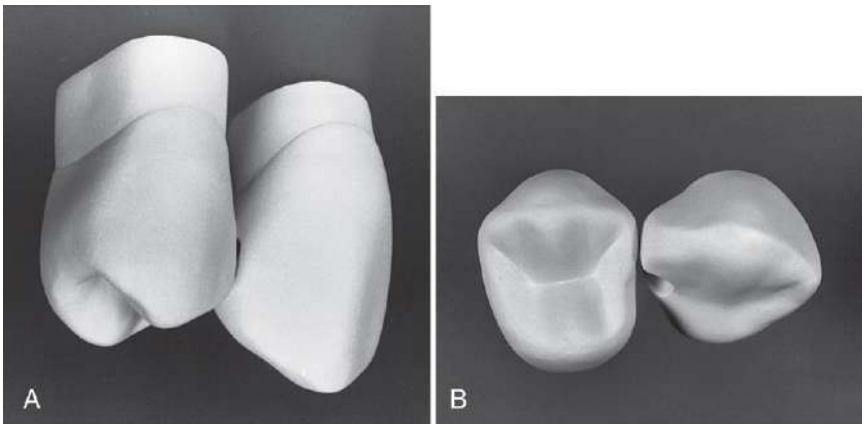
When preparing a gingival wall that is near the level of the rubber dam or apical to it, it is beneficial to place a wedge in the gingival embrasure earlier to depress and protect soft tissue and the rubber dam. As the bur is preparing the gingival wall, it may lightly shave the wedge. A triangular (i.e. anatomic) wedge, rather than a round wedge, is used for a deep gingival margin.

The initial tooth preparation is completed by using a No. ½ round bur to accentuate the axial line angles (Online Fig. 26.12A and B), particularly the axiogingival angle. This facilitates the subsequent placement of retention grooves and leaves the internal line angles slightly rounded. Rounded internal preparation angles permit more complete condensation of the amalgam. The No. ½ round bur may also be used to smooth any roughened, undermined enamel produced at the gingival and facial cavosurface margins (Online Fig. 26.12C). The incisal margin of the minimally extended preparation is often not accessible to the larger round bur without marring the adjacent tooth (Online Fig. 26.12D). Further finishing of the incisal margin is presented later. At this point, the initial tooth preparation is completed.

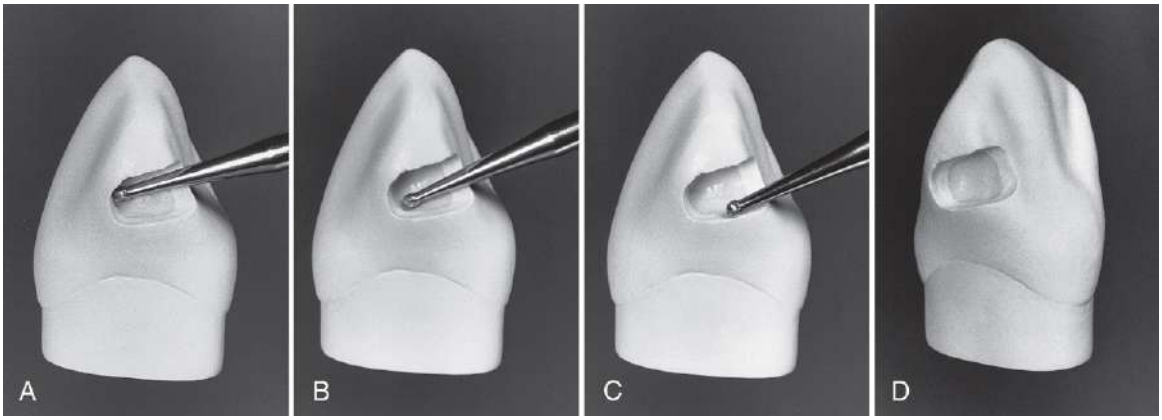
Final Tooth Preparation

Final tooth preparation involves removing any remaining infected dentin; protecting the pulp; developing secondary resistance and retention forms; finishing external walls; and cleaning, inspecting and desensitizing or bonding. Any remaining infected carious dentin on the axial wall is removed by using a slowly revolving round bur (No. 2 or No. 4), appropriate spoon excavators, or both.

For the class III amalgam restoration, resistance form against post-restorative fracture is provided by (i) cavosurface and amalgam margins of 90 degrees, (ii) enamel walls supported by sound dentin, (iii) sufficient bulk of amalgam (minimal 1-mm thickness) and (iv) no sharp preparation internal angles. The box-like preparation form provides primary retention form. Secondary retention form is provided by a gingival groove, an incisal cove and sometimes a lingual dovetail.



Online Figure 26.11 Distofacial (A) and incisal (B) views of the canine to show the curved proximal outline necessary to preserve the distoincisal corner of the tooth. The incisal margin of this preparation example is located slightly incisally of the proximal contact (but whenever possible, the margin may be in the contact area).



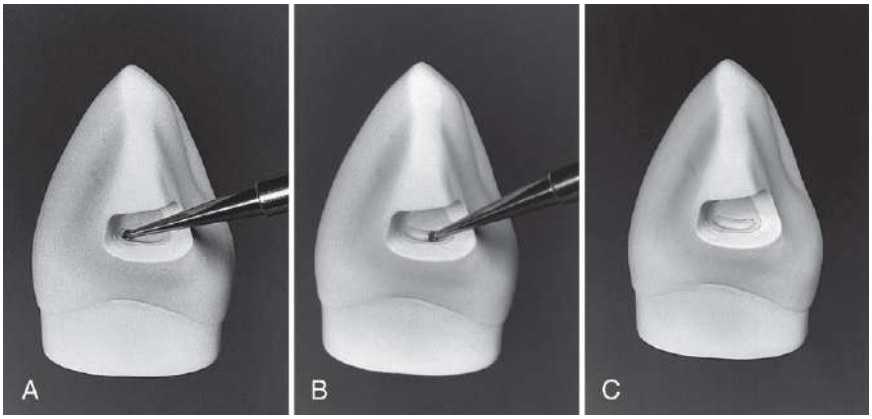
Online Figure 26.12 Refining proximal portion. (A–C) A small, round bur is used to shape the preparation walls, define line angles and initiate removal of any undermined enamel along the gingival and facial margins. (D) Tooth preparation completed, except for the final finishing of the enamel margins and placing the retention form.

The gingival retention groove is prepared by placing a No. ¼ round bur (rotating at low speed) in the axio-facio-gingival point angle. It is positioned in the dentin to maintain 0.2 mm of dentin between the groove and the DEJ. The rotating bur is moved lingually along the axiogingival line angle, with the angle of cutting generally bisecting the angle between the gingival and axial walls. Ideally, the direction of the gingival groove is slightly more gingival than axial (and the direction of an incisal [i.e. occlusal] groove would be slightly more incisal [i.e. occlusal] than axial) (Online Fig. 26.13; see also Fig. 26.8).

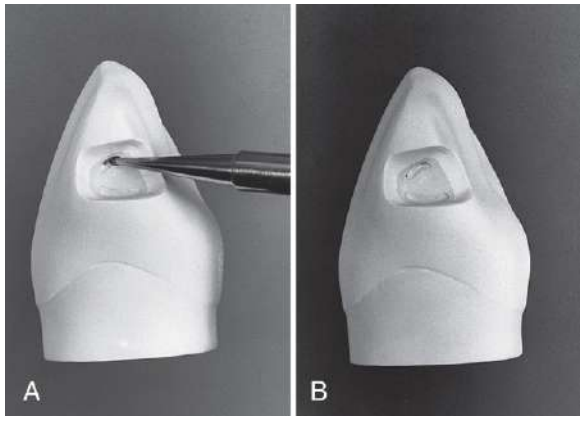
Alternatively, if less retention form is needed, two gingival coves may be used, as opposed to a continuous groove. One each may be placed in the axio-gingivo-facial and axio-gingivo-lingual point angles. The diameter of the ¼ round bur is 0.5 mm, and the depth of the groove should be half this diameter (0.25 mm). (See the location and depthwise direction of the groove, where the gingival wall remains in enamel, in Fig. 26.18.) When preparing a retention groove on the root surface (gingival wall in cementum or dentin), the angle of cutting is more gingival, resulting in the distance from the gingival cavosurface margin to the groove being approximately 0.3 mm (Online Fig. 26.8). Careful technique is necessary in preparing the gingival retention groove. If the dentin that supports gingival enamel is removed, enamel is subject to fracture. In addition, if the groove is placed only in the axial wall, no effective retention form is developed, and a risk of pulpal involvement is possible.

An incisal retention cove is prepared at the axio-facio-incisal point angle with a No. ¼ round bur in dentin, being careful not to undermine enamel. It is directed similarly into the incisal point angle and prepared to half the diameter of the bur (Online Fig. 26.14). Undermining the incisal enamel should be avoided. For the maxillary canine, the palm-and-thumb grasp may be used to direct the bur incisally (Online Fig. 26.15). This completes the typical class III amalgam tooth preparation (Online Fig. 26.16). Similar to class I and II amalgams, it is recommended that the clinician prepares mechanical retention.

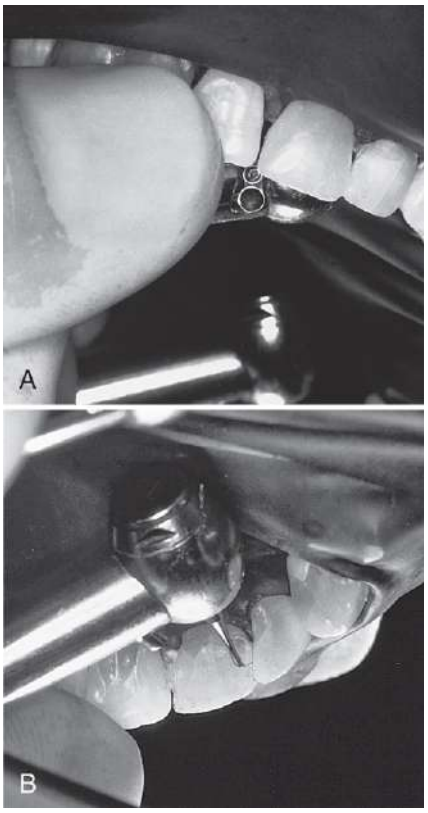
A lingual dovetail is not required in small or moderately sized class III amalgam restorations. It may be used in large preparations, especially preparations with excessive incisal extension in which additional retention form is needed.



Online Figure 26.13 Preparing the gingival retention form. (A) Position of No. ¼ round bur in axio-facio-gingival point angle. (B) Advancing the bur lingually to prepare the groove along the axiogingival line angle. (See Online Fig. 26.8 regarding location, depth direction and direction depth of groove.) (C) Completed gingival retention groove.



Online Figure 26.14 Preparing the incisal retention cove. (A) Position of No. ¼ round bur in the axioincisal point angle. (B) Completed incisal cove.



Online Figure 26.15 Use of the palm-and-thumb grasp to place the incisal retention cove. (A) Hand position showing thumb rest. (B) Hand-piece position for preparing the incisal retention.

The dovetail may not be necessary (even in large preparations), however, if an incisal secondary retention form can be accomplished (Online Fig. 26.17).

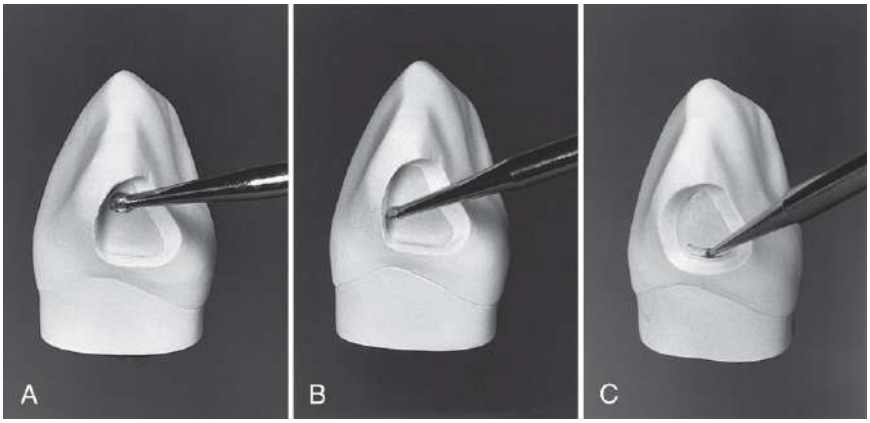
If a lingual dovetail is needed, it is prepared only after initial preparation of the proximal portion has been completed. Otherwise, the tooth structure needed for the isthmus between the proximal portion and the dovetail may be removed when the proximal outline form is prepared. The lingual dovetail should be conservative, generally not extending beyond the mesiodistal midpoint of the lingual



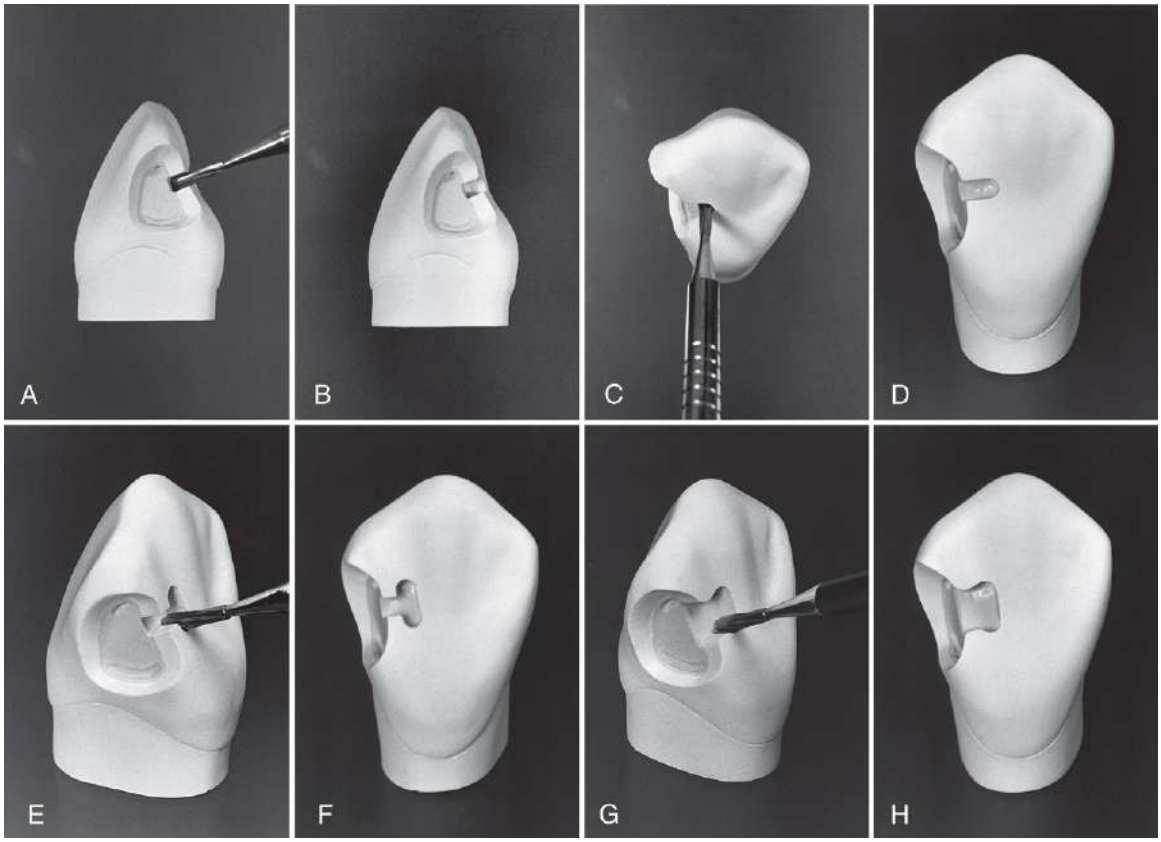
Online Figure 26.16 Completed class III tooth preparation for amalgam restoration.

surface; this varies according to the extent of the proximal caries. The axial depth of the dovetail should approximate 1 mm, and the axial wall should be parallel to the lingual surface of the tooth. This wall may or may not be in dentin. The No. 245 bur is positioned in the proximal portion at the correct depth and angulation and moved in a mesial direction (Online Fig. 26.18A and B). The correct angulation places the long axis of the bur perpendicular to the lingual surface. The bur is moved to the point that corresponds to the most mesial extent of the dovetail (Online Fig. 26.18C and D). The bur is then moved incisally and gingivally to create sufficient incisogingival dimension to the dovetail (approximately 2.5 mm) (Online Fig. 26.18E and F). The incisal and gingival walls of the isthmus are prepared in smooth curves connecting the dovetail to the proximal outline form (Online Fig. 26.18G and H).

The gingival margin trimmer can be used to bevel (or round) the axiopulpal line angle (i.e. the junction of the proximal and dovetail preparation). This increases the strength of the restoration at the junction of the proximal and lingual portions by providing bulk and reducing stress concentration. The lingual convergence of the dovetail's external walls (prepared with the No. 245 bur) usually provides a sufficient retention form. Retention coves, one in the incisal corner and one in the gingival corner (Online Fig. 26.19), may be placed in the dovetail to enhance retention if the axial wall of the dovetail is in dentin. The coves are prepared with the No. ¼ round bur in dentin that does not immediately support the lingual enamel. This preparation may require deepening of the axial wall. Unsupported enamel is removed, the walls or margins are smoothed and the cavosurface angles are refined, where indicated. The 8-3-22 hoe is recommended for finishing minimally extended margins (Online Fig. 26.20). If the gingival margin is in enamel, a slight bevel (approximately 20 degrees) is necessary to ensure full-length enamel rods forming the cavosurface margin. All the walls of the preparation should meet the external tooth surface to form a right angle (i.e. butt joint) (Online Fig. 26.21; see also Fig. 26.16). The various steps involved



Online Figure 26.17 Extensive class III tooth preparation. (A) Initial tooth preparation with No. 2 round bur. (B) Defining line angles and removing undermined enamel with No. ½ round bur. (C) Placing the retention groove using No. ¼ round bur. Note the completed incisal cove.



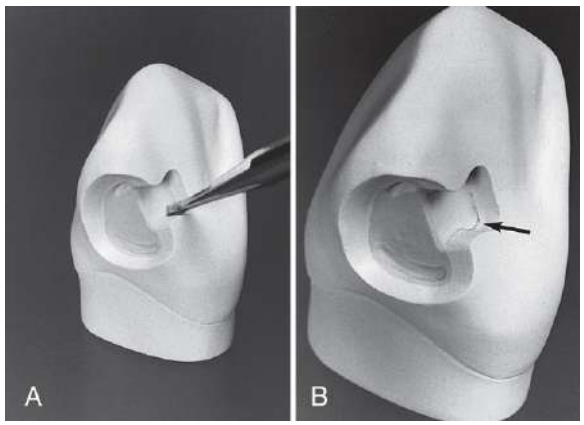
Online Figure 26.18 Lingual dovetail providing additional retention for extensive amalgam restoration. (A) Bur position at correct depth and angulation to begin cutting. (B) Initial cut in beginning dovetail. (C) Bur moved to most mesial extent of dovetail. (D) If possible, cutting should not extend beyond the midlingual position. (E) Bur cutting gingival extension of the dovetail. (F) Incisal and gingival extensions of the dovetail. (G) Completing the isthmus. The proximal and lingual portions are connected by the incisal and gingival walls in smooth curves. (H) Completed lingual dovetail.

in the clinical procedure with the dovetail are shown in Online Fig. 26.22. The completed tooth preparation should be cleaned of any residual debris and inspected. Careful assessment should be made to see that all of the caries has been removed, that the depth and retention are appropriate and that cavosurface margins provide for the amalgam bulk.

Restorative Technique

Desensitizer Placement

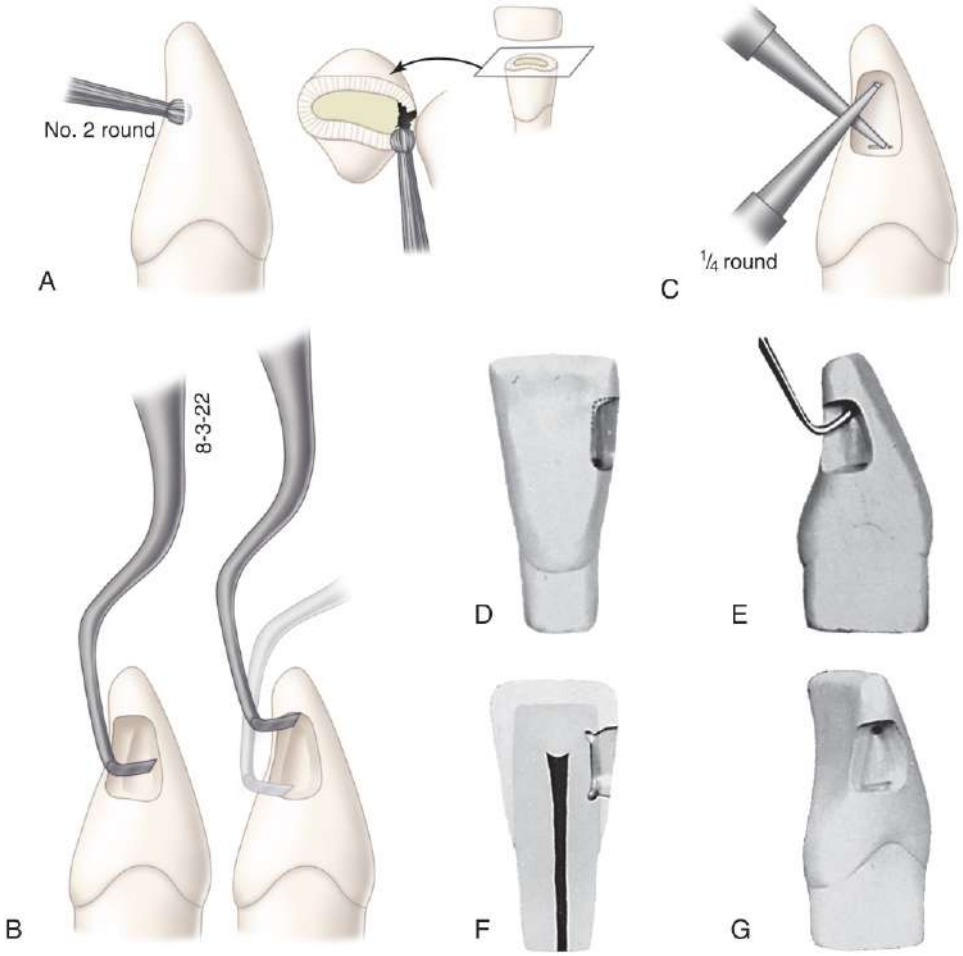
The use of a dentin desensitizer over the prepared tooth structure before placing amalgam is generally recommended. The dentin desensitizer is rubbed onto the prepared



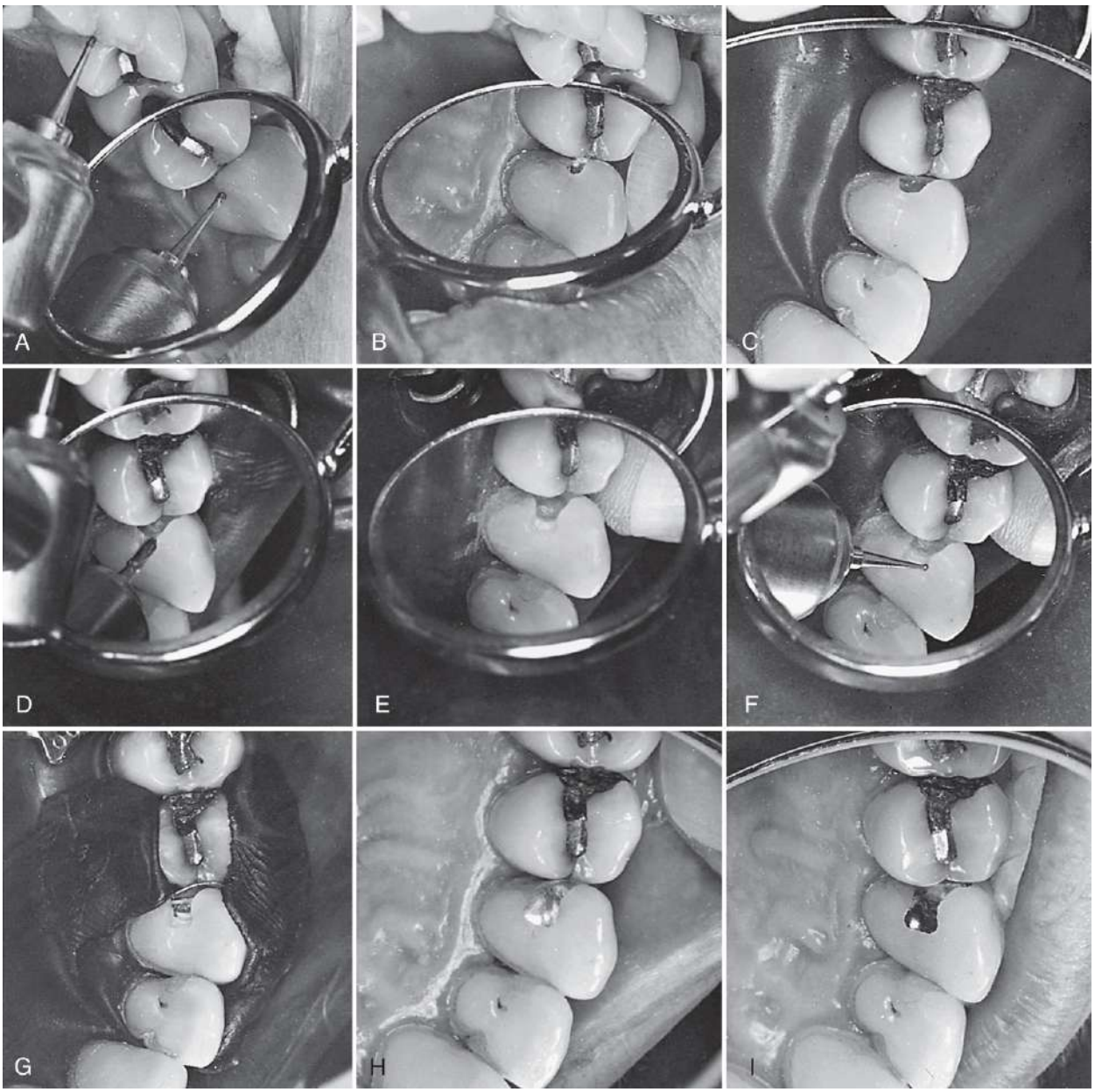
Online Figure 26.19 Ensuring retention in lingual dovetail (often optional). (A) Position of No. 33½ bur for cutting the retention cove. (B) Preparation of the cove should not remove the dentinal support of the lingual enamel *(arrow)*.



Online Figure 26.21 Completed distolingual class III tooth preparation for amalgam.



Online Figure 26.20 Class III tooth preparation for amalgam restoration on the mandibular incisor. (A) Entering the tooth from the lingual approach. (B) Finishing the facial, incisal and gingival enamel margins with an 8-3-22 triple angle hoe. Note how the reverse bevel blade is used on the gingival enamel. (C) Placing incisal and gingival retention forms with No. ¼ round bur. (D) *Dotted line* indicates the outline of the additional extension that is sometimes necessary for access in placing the incisal retention cove. (E) Position of a bi-bevelled hatchet to place the incisal retention cove. (F) The axial wall forms a convex surface over the pulp. (G) Completed tooth preparation. Note the gingival retention groove.

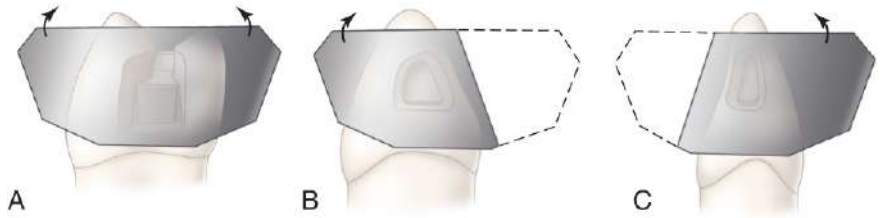


Online Figure 26.22 Distolingual tooth preparation and restoration. (A) Bur position for entry. (B) Penetration made through the lingual enamel to the caries. (C) Proximal portion completed, except for the retention form. (D) Preparing the dovetail. (E) Completed preparation, except for the retention groove and the coves. (F) Bur position for the incisal cove in the dovetail. (G) Rigid material-supported matrix ready for the insertion of amalgam. (H) Carving completed and rubber dam removed. (I) Polished restoration.

tooth surface for 30 s and excess moisture is removed without desiccating the dentin.

Matrix Placement

The wedged, rigid material-supported sectional matrix may be used for the class III amalgam restoration. Insertion of amalgam into the class III tooth preparation is usually from the lingual aspect. It is essential to trim the lingual portion of the sectional matrix material correctly to avoid covering the preparation and blocking access for insertion of the amalgam. A length of 5y16 inch (8 mm) wide, 0.002 inch (0.05 mm) thick stainless steel matrix material that covers one-third of the facial surface and extends through the proximal to the lingual surface is obtained. The lingual portion is trimmed at an angle that corresponds approximately to the slope of the lingual surface of the tooth (Online Fig. 26.23). The section of matrix is burnished on a resilient paper pad to create the desired contact and contour form. The sectional matrix is placed in position and wedged from the facial or lingual embrasure, whichever is greater. The facial and lingual portions of the matrix are stabilized with a rigid material (Online Fig. 26.22G). Precontoured metallic matrices may be used (instead of custom-made matrices) if the contour of the precontoured matrix coincides with that of the proximal surface being



Online Figure 26.23 Matrix strip design. (A) Design required for rigid material-supported matrix for class II tooth preparations. (B) Alteration necessary for class III preparation on the maxillary canine. (C) Alteration necessary for the mandibular incisor. The strip material is cut to approximate the slope of the lingual surface.

restored. If the preparation is small and the matrix is sufficiently rigid, additional rigid material to support the matrix may not be required.

Condensation and Carving

Insertion of the amalgam, initial carving, matrix removal, wedge removal and final carving are similar to the techniques for posterior teeth. When properly placed, conservative restorations in incisors and canines are relatively inconspicuous (Online Fig. 26.24). Online Fig. 26.25 illustrates a class III amalgam restoration in a mandibular incisor.

Finishing and Polishing

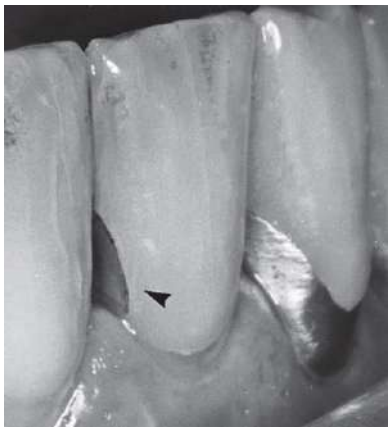
The finishing and polishing techniques and procedures are the same as those presented in Chapter 13.

Clinical Technique for Class V Amalgam Restorations

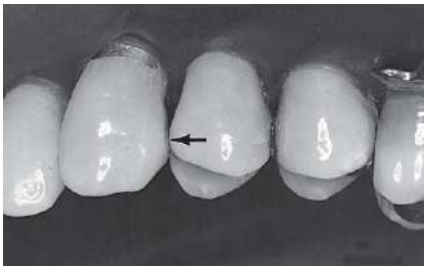
Initial Procedures

Proper isolation prevents moisture contamination of the operating site, enhances asepsis and facilitates access and visibility. Moisture in the form of saliva, gingival sulcular fluid or gingival haemorrhage must be excluded during caries removal, liner application, and insertion and carving of amalgam. Moisture impairs visual assessment, may contaminate the pulp during caries removal (especially with a pulpal exposure) and negatively affects the physical properties of restorative materials. The gingival margin of class V tooth preparations is often apical to the gingival crest. Such a gingival margin necessitates retraction of the free gingiva with a retraction cord or appropriate rubber dam and retainer to protect it and to provide access while eliminating seepage of sulcular fluid into the tooth preparation or restorative materials.

These isolation objectives are met by local anaesthesia and isolation by (i) a cotton roll and a retraction cord or (ii) a rubber dam and a suitable cervical retainer (Online Fig. 26.26). Isolation with a cotton roll and a retraction cord is satisfactory when properly performed. This type of isolation is practical and probably the approach most often used. The retraction cord should be placed in the sulcus before initial tooth preparation to reduce the possibility of cutting instruments damaging the free gingiva. The cord



Online Figure 26.25 Class III amalgam restoration on the mandibular incisor *(arrowhead)*.



Online Figure 26.24 Inconspicuous facial margin *(arrow)* of class III amalgam restoration on the maxillary canine.



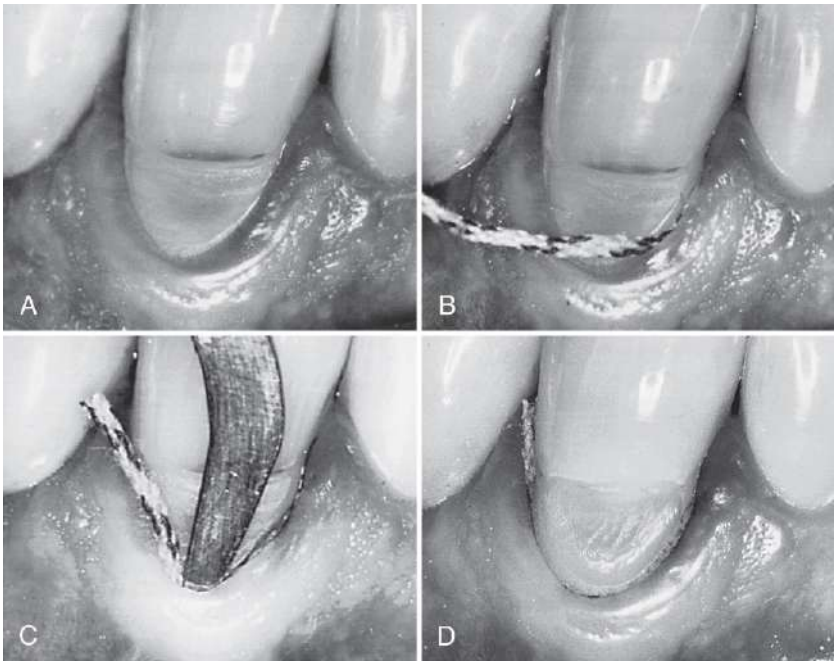
Online Figure 26.26 A rubber dam and a No. 212 retainer may be required to isolate the carious area properly.

should produce a temporary, adequate, a traumatic apical retraction and lateral deflection of the free gingiva. The cord may be treated with haemostatic preparations containing aluminium chloride or ferrous sulphate. Alternatively, the cord may be treated with epinephrine. However, caution must be used as epinephrine on abraded gingiva can be absorbed rapidly into the circulatory system, causing an increase in blood pressure, elevated heart rate and possible dysrhythmia. The retraction cord may be braided, twisted or woven. The diameter of the cord should be easily accommodated in the gingival sulcus. An appropriate amount is cut to a length ¼ inch (6 mm) longer than the gingival margin. Some operators prefer to place the cord in a Dappen dish, wet it with a drop of haemostatic solution and blot it with a 2 × 2 inch (5 × 5 cm) gauze to remove excess liquid. A braided or woven cord is usually easier to use because it does not unravel during placement. A larger cord can be inserted over the first cord if the sulcus is large enough to accommodate two cords. A thin, blunt-edged instrument blade or the side of an explorer tine may be used to gently insert the cord progressively into place. A slight backward direction of the instrument as it steps along the cord helps prevent dislodgment of previously inserted cord (Online Fig. 26.27). In addition, using a second instrument stepping along behind the first instrument can help prevent dislodgment of cord. In addition, using the air syringe or cotton pellets to reduce or absorb the sulcular fluid in the cord already placed is helpful during cord placement. The cord results in adequate retraction in a short time. If significant blanching of the free gingiva is observed (or if too much pressure has to be applied to place the cord), it means that an oversized cord has been selected, and it should therefore be exchanged with a cord of smaller diameter. The cord can be moistened before or after placement with the haemostatic solution if slight haemorrhage is anticipated or observed. Alternatively, the cord can be used dry.

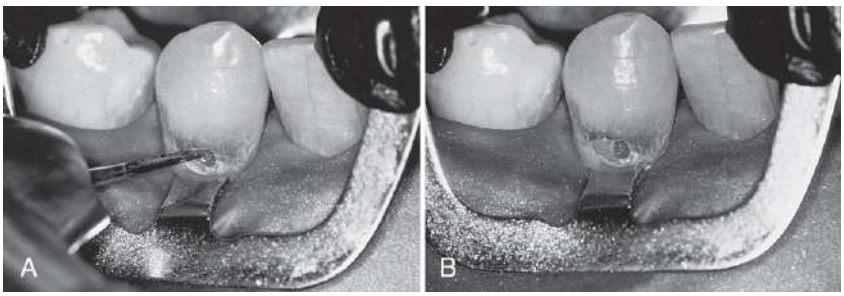
The cord usually remains in place throughout tooth preparation as well as insertion and initial carving of amalgam. While carving amalgam at the gingival margin, the presence of the cord may cause difficulty in feeling the unprepared tooth surface apical to the margin to prevent undercarving of the margin; this results in overcontouring and marginal excess. In this instance, after carving gross excess, the cord can be teased from its place before the carving is completed.

Tooth Preparation

Proper outline form for class V amalgam tooth preparations results in extending the cavosurface margins to sound tooth structure while maintaining a limited axial depth of 0.5 mm inside the DEJ and 0.75 mm inside cementum (when on the root surface). The outline form for the class V amalgam tooth preparation is determined primarily by the location and size of the caries or old restorative material. Clinical judgement determines final preparation outline, especially when the cavosurface margins approach or extend into areas of enamel decalcification. The operator must observe the prepared enamel wall to evaluate the depth of the decalcified enamel and to determine if cavitation exists peripheral to the wall. When no cavitation has occurred and when the decalcification does not extend appreciably into the enamel, extension of the outline form often should cease. In some cases, if all decalcification



Online Figure 26.27 Use of the retraction cord for isolation of a class V lesion. (A) Pre-operative view. (B) Cord placement initiated. (C) Cord placement using a thin, flat-bladed instrument. (D) Cord placement completed.

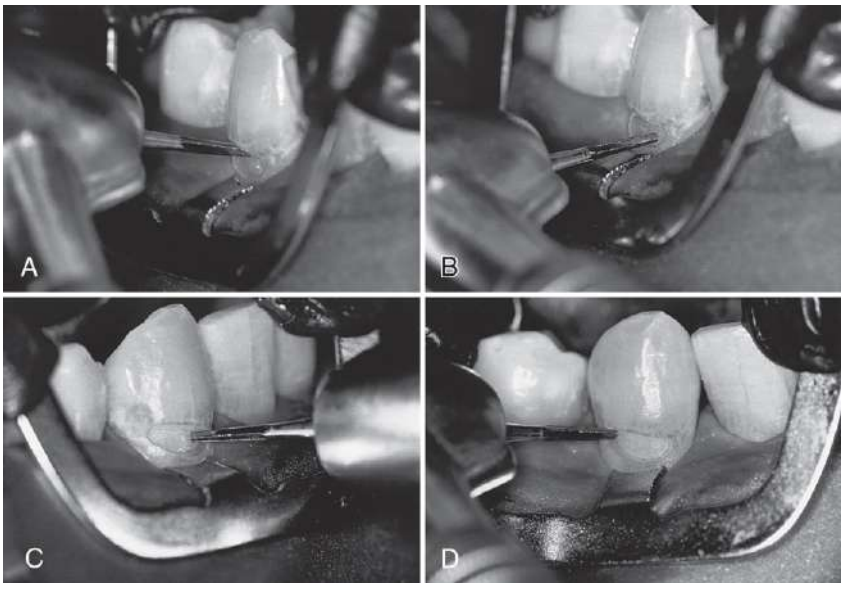


Online Figure 26.28 Starting class V tooth preparation. (A) Bur positioned for entry into caries lesion. (B) The entry cut is the beginning of the outline form having a limited axial depth. (The end of the bur in the centre of the lesion may be in the carious tooth structure or in the air.)

were included in the outline form, the preparation would extend into the proximal cervical areas (if not circumferentially around the tooth). Such a preparation would be difficult and perhaps unrestorable. A full-coverage restoration should be considered for teeth with extensive cervical decalcification.

Initial Tooth Preparation

A class V amalgam restoration is not used often in a mandibular canine, but it is presented here for illustration. The same general principles for tooth preparation apply for all other tooth locations. A tapered fissure bur of suitable size (e.g. No. 271) is used to enter the caries lesion (or existing restoration) to a limited initial axial depth of 0.5 mm inside the DEJ (Online Fig. 26.28). This depth is usually 1–1.25 mm total axial depth, depending on the incisogingival (i.e. occlusogingival) location. The enamel is considerably thicker occlusally and incisally than cervically. If the preparation is on the root surface, however, the axial depth is approximately 0.75 mm. The end of the bur at the initial depth is in dentin, in infected carious dentin or in old restorative material. The edge of the end of the bur can be used to penetrate the area; this is more efficient than using the flat end of the bur, reducing the possibility of the bur's 'crawling'. When the entry is made, the bur orientation is adjusted to ensure that all external walls are perpendicular to the external tooth surface and parallel to the enamel rods (Online Fig. 26.29). Often, this requires changing the orientation of the hand-piece to accommodate the cervical mesiodistal and incisogingival (i.e. occlusogingival) convexity of the tooth. The preparation is extended incisally, gingivally, mesially and distally until the cavosurface margins are positioned in sound tooth structure such that an initial axial depth of 0.5 mm inside the DEJ (if on the root surface, the axial depth is 0.75 mm) is established. When extending mesially and distally, it may be necessary to protect the rubber dam from the bur by placing a flat-bladed instrument over the dam (Online Fig. 26.30). Because the axial wall follows the mesiodistal and incisogingival (i.e. occlusogingival) contours of the facial surface of the tooth, it is usually convex in both directions. In addition, the axial wall usually is slightly deeper at the incisal wall, where more enamel (i.e. approximately 1–1.25 mm in depth) is



Online Figure 26.29 When extending incisally (A), gingivally (B), mesially (C) and distally (D), the bur is positioned to prepare these walls perpendicular to the external tooth surface.



Online Figure 26.30 The flat-bladed instrument protects the rubber dam from the bur.

present than at the gingival wall, where little or no enamel (i.e. approximately 0.75–1 mm in depth) may be present. A depth of 0.5 mm inside the DEJ permits placement of necessary retention grooves without undermining enamel. This subtle difference in depth also serves to increase the thickness of the remaining dentin (between the axial wall and the pulp) in the gingival aspect of the preparation to aid in protecting the pulp. For the tooth preparation that is extended incisogingivally, the axial wall should be more convex (because it follows the contour of the DEJ).

Alternatively, an appropriate carbide bur (usually No. 2 or No. 4) may be used for the initial tooth preparation. Round burs are indicated in areas inaccessible to a fissure bur that is held perpendicular to the tooth surface. If needed, smaller round burs may also be used to define the internal angles in these preparations, enhancing proper placement of the retention grooves.

Final Tooth Preparation

Final tooth preparation involves removing any remaining infected dentin; pulp protection; retention form; finishing external walls; and cleaning, inspecting and desensitizing. Any remaining infected axial wall dentin is removed with a No. 2 or No. 4 round bur. Any old restorative material (including base and liner) remaining may be left if (i) no clinical or radiographic evidence of recurrent caries exists, (ii) the periphery of the base and liner is intact and (iii) the tooth is asymptomatic. With proper outline form, the axial line angles are already in sound dentin. If needed, an appropriate liner or base is applied.

Because the mesial, distal, gingival and incisal walls of the tooth preparation are perpendicular to the external tooth surface, they usually diverge facially. Consequently, this form provides no inherent retention, and retention form must be provided because the primary retention form for an amalgam restoration is macromechanical. A No. ¼ round bur is used to prepare two retention grooves, one along the incisoaxial line angle and the other along the gingivoaxial line angle (Online Fig. 26.31). The handpiece is positioned so that the No. ¼ round bur is directed generally to bisect the angle formed at the junction of the axial wall and the incisal (i.e. occlusal) wall. Ideally, the direction of the incisal (i.e. occlusal) groove is slightly more incisal (i.e. occlusal) than axial, and the direction of the gingival groove is slightly more gingival than axial. Alternatively, four retention coves may be prepared, one in each of the four axial point angles of the preparation (Online Fig. 26.32).

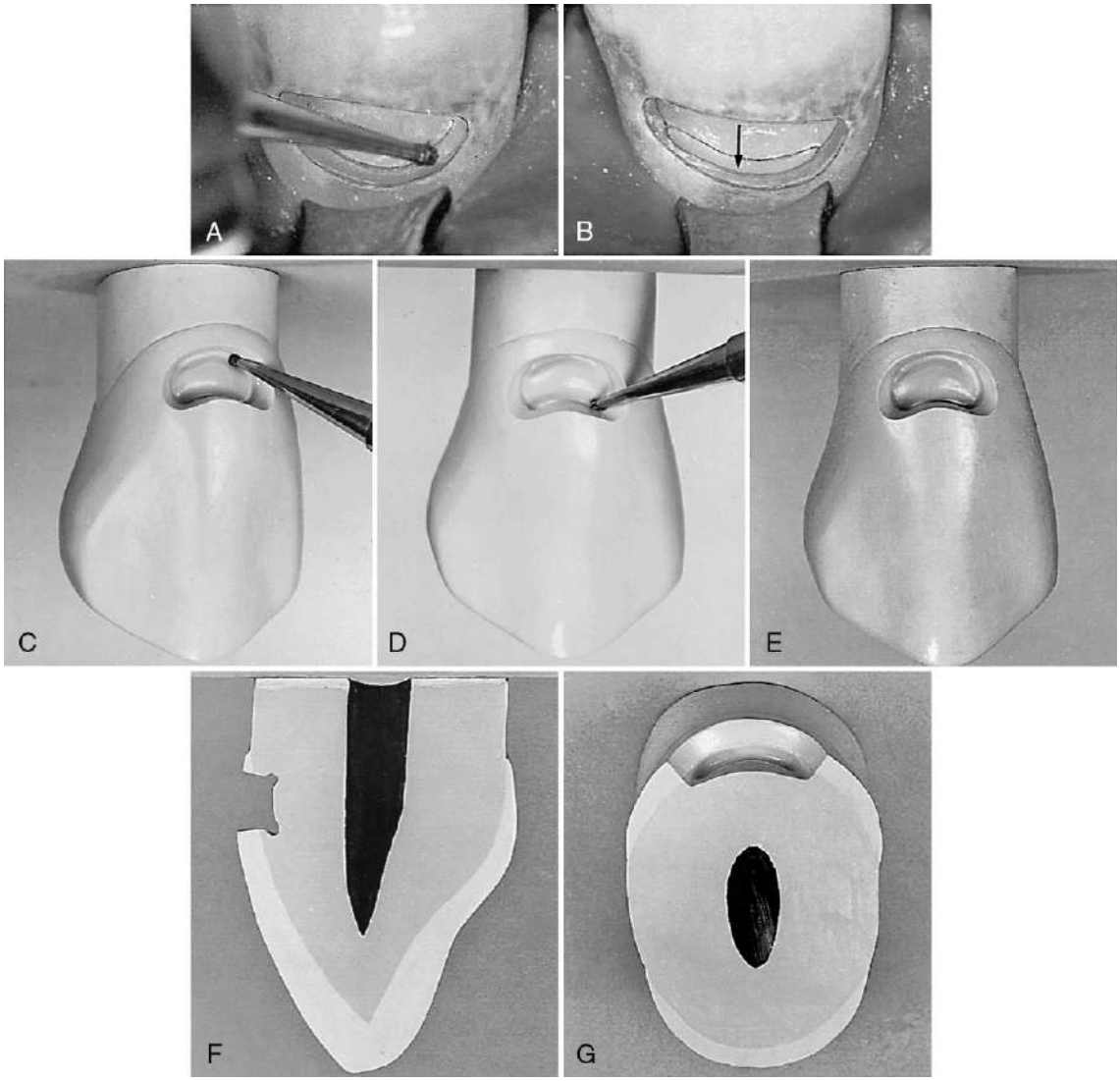
Using four coves instead of two full-length grooves conserves the dentin near the pulp, reducing the possibility of a mechanical pulp exposure. The depth of the grooves should be approximately 0.25 mm, which is half the diameter of the bur. It is important that the retention grooves be adequate because they provide the only retention form to the preparation. Regardless, the grooves should not remove dentin immediately supporting enamel. In a large class V amalgam preparation, extending the retention groove circumferentially around all the internal line angles of the tooth preparation may enhance the retention form.

If access is inadequate for use of the No. ¼ round bur, an angle-former chisel may be used to prepare the retention form. In addition, a No. 33½ bur can be used. Both methods result in retention grooves that are angular, but positioned in the same location and approximately to the same depth as when the No. ¼ round bur is used. The rounded retention form placed with the No. ¼ round bur is generally preferred, however, because amalgam can be condensed into rounded areas better than into sharp areas, resulting in better adaptation of amalgam into the retention grooves. If necessary, suitable hand instruments (e.g. chisels, margin trimmers) are used to plane the enamel margins, verifying soundness and 90-degree cavosurface angles. Finally, the preparation is cleaned and inspected for completeness. A desensitizer is then applied.

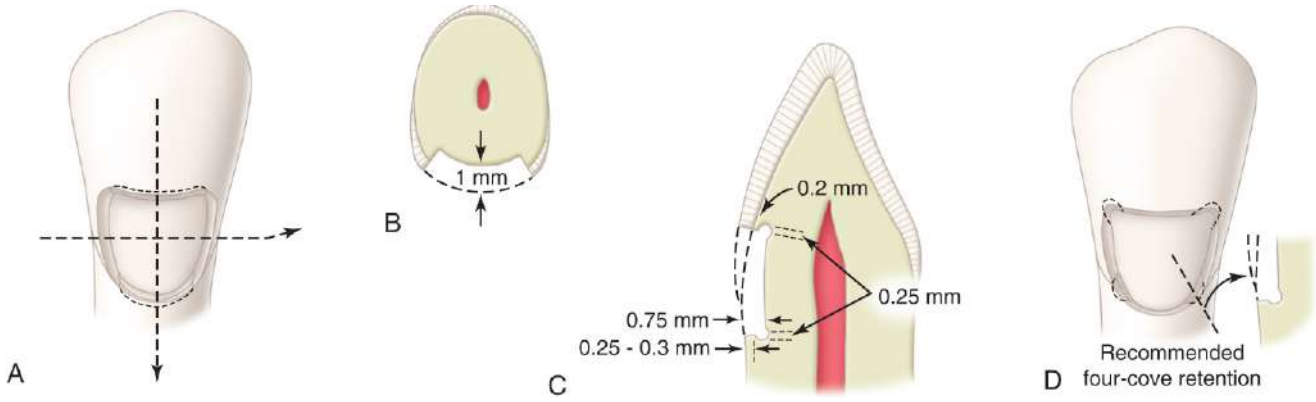
Large Preparations That Include Line Angles

Caries on the facial (or lingual) surface may extend beyond the line angles of the tooth. Maxillary molars, particularly the second molars, are most commonly affected by these extensive defects (Online Fig. 26.33A). In this example, if the remainder of the distal surface is sound and the distal caries is accessible facially, the facial restoration should extend around the line angle. This prevents the need for a class II proximal restoration to restore the distal surface. As much of the preparation as possible should be completed with a fissure bur. A round bur, approximately the same diameter as the fissure bur, is then used to initiate the distal portion of the preparation (Online Fig. 26.33B and C). Smaller round burs should be used to accentuate the internal line angles of the distal portion. Preparing the facial portion first provides better access and visibility to the distal portion. Occasionally, hand instruments may be useful for completing the distal half of the preparation when space for the handpiece is limited (Online Fig. 26.33D–F).

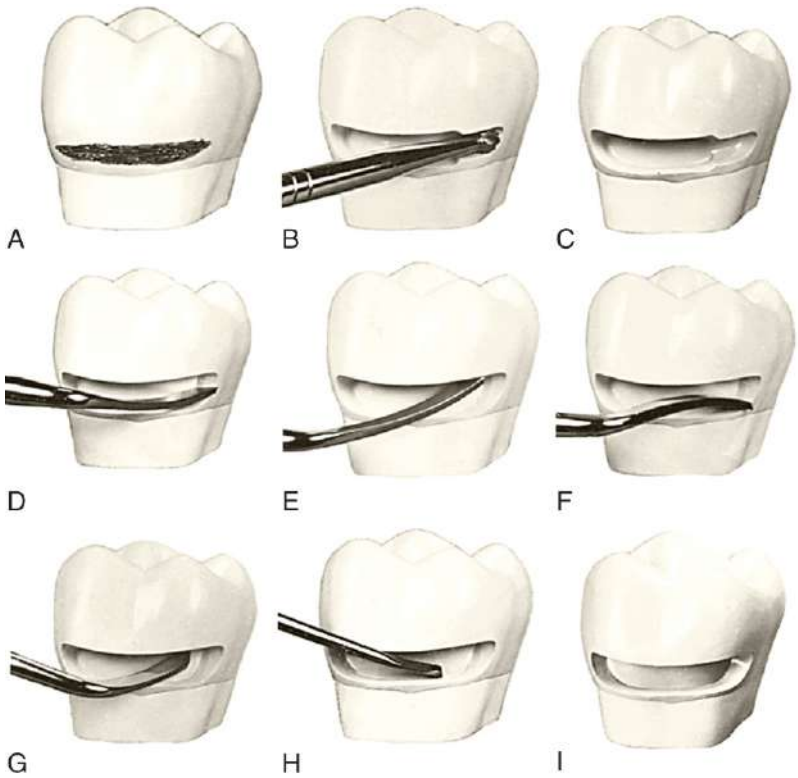
Grooves placed along the entire length of the occlusoaxial and gingivoaxial line angles help ensure retention of



Online Figure 26.31 Retention form. (A) A No. ¼ round bur positioned to prepare the gingival retention groove. (B) Gingival retention groove *(arrow)* prepared along the gingivoaxial line angle generally to bisect the angle formed by the gingival and axial walls. Ideally, the direction of preparation is slightly more gingival than pulpal. An incisal retention groove is prepared along the incisoaxial line angle and directed similarly. (C, D) A groove is placed with a No. ¼ round bur along the gingivoaxial and incisoaxial line angles 0.2 mm inside the dentinoenamel junction (DEJ) and 0.25 mm deep. Note the slight pulpal inclination of the shank of the No. ¼ round bur. (E) Facial view. (F) Incisogingival section. Grooves depthwise are directed mostly incisally (gingivally) and slightly pulpally. (G) Mesiodistal section.



Online Figure 26.32 (A–C) Extended class V tooth preparation (A) with the axial wall contoured parallel to the dentinoenamel junction (DEJ) mesiodistally (B) and incisogingivally (C). The axial wall pulpal depth is 1 mm in the crown and 0.75 mm in the root. In addition, note location and direction depth (0.25 mm) of the retention grooves and the dimension of the gingival wall (0.25 mm) from the root surface to the retention groove. (D) Large class V preparation with retention coves prepared in the four axial point angles.



Online Figure 26.33 Tooth preparation on maxillary molar. (A) Caries extending around distofacial corner of the tooth. (B, C) Distal extension is accomplished with round bur. (D–F) A gingival margin trimmer may be useful in completing the distal half of the preparation when hand-piece access is limited. (G) A gingival margin trimmer may be used to provide the retention grooves. (H) An angle-former chisel may be used to prepare the retention grooves in the distal portion of the preparation. (I) Completed tooth preparation.

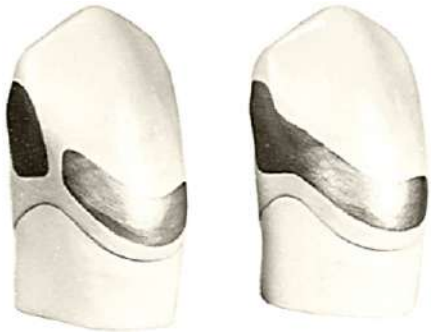
the restoration. The No. ¼ round bur is used as previously described to prepare the retention grooves. A gingival margin trimmer or a 7-85-2½-6 angle-former chisel can be used in the distal half of the preparation to provide retention form when access for the handpiece is limited (Online Fig. 26.33G and H).

Because of the proximity of the coronoid process, access to the facial surfaces of maxillary molars, particularly the second molars, is often limited. Having the patient partially closes and shifts the mandible towards the tooth being restored improves access and visibility (Online Fig. 26.34).

If the class V outline form approaches an existing proximal restoration, it is better to extend slightly into the bulk of the proximal restoration, rather than to leave a thin section of the tooth structure between the two restorations (Online Fig. 26.35). In this illustration, the previously placed amalgam served as the distal wall of the preparation. When proper treatment requires class II and V amalgam restorations on the same tooth, the class II preparation and restoration is completed before initiating the class V restoration. If the class V restoration were done first, it might be damaged by the matrix band and wedge needed for the class II restoration.



Online Figure 26.34 The mandible shifted laterally for improved access and visibility.



Online Figure 26.35 When a class V outline form closely approaches an existing restoration, the preparation should be extended to remove the remaining thin enamel wall to achieve adjoining restorations.

Restorative Technique

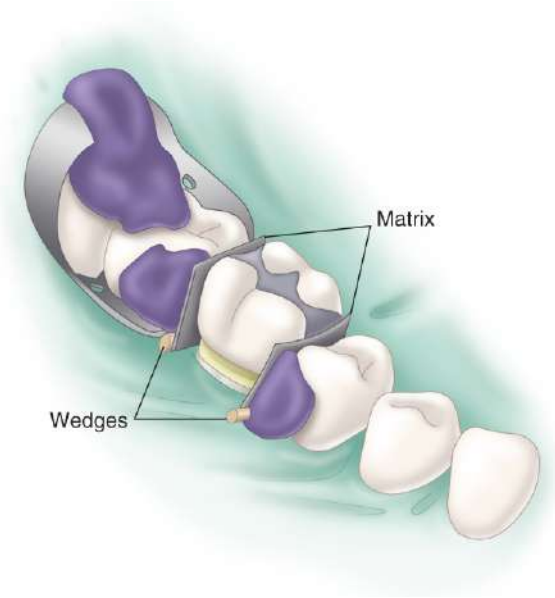
Desensitizer Placement

The same considerations presented earlier apply for the class V amalgam restoration.

Matrix Placement

Most class V amalgam restorations are placed without the use of any type of matrix. The most difficult condensation occurs in a tooth preparation with an axial wall that is convex mesiodistally. Two alternative methods for insertion may be used. The preferred method is the application of a matrix that confines amalgam in the mesial and distal portions of the preparation (Online Fig. 26.36). Short lengths of stainless steel matrix material, one each for the mesial and distal surfaces, are passed through the proximal contacts, carefully guided into the gingival sulcus and wedged. The strips must be wide enough to extend occlusally through the respective proximal contacts and long enough to extend slightly past the facial line angles. The strip usually requires rigid material support for stability. The strips offer resistance against condensing the mesial and distal portions, which provides support for condensing the centre of the restoration. The gingival edge of the steel strip often must be trimmed to conform to the circumferential contour (level) of the base of the gingival sulcus to prevent soft tissue damage. Rather than using two short pieces, the operator can use a longer length that may be passed through one proximal contact, extended around the lingual surface and passed through the other contact, forming a U-shaped matrix. Trimming the gingival edge to conform to the interproximal soft tissue anatomy usually is more difficult with one matrix strip than when two strips are used.

A conventional Tofflemire band and retainer may be used with a window cut into the band allowing access to the preparation for condensation (Online Fig. 26.37). Alternatively, the tooth may be prepared and restored in sections without using a matrix. Each successive section of the preparation should be extended slightly into the previously condensed portion to ensure caries removal. This procedure is time-consuming but effective.

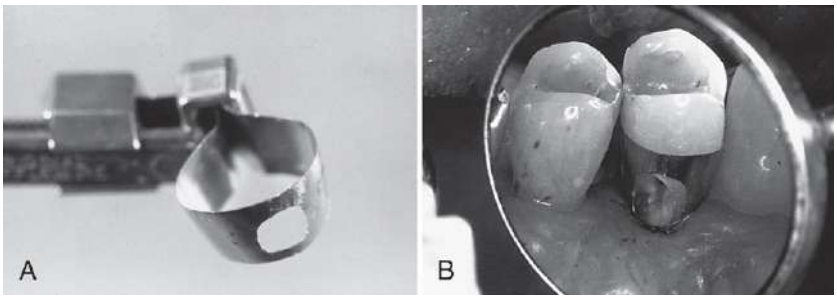


Online Figure 26.36 Application of the matrix to confine amalgam in the mesial and distal extensions of the preparation.

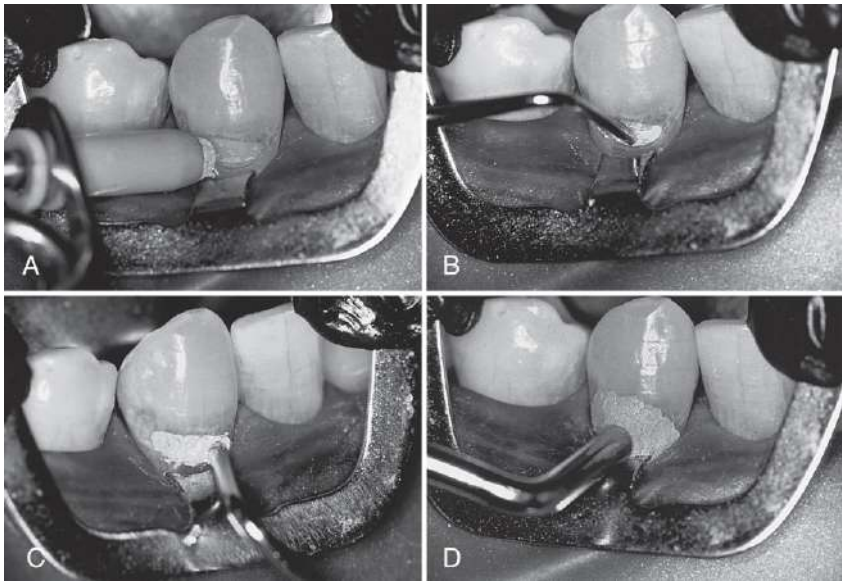
Insertion and Carving of the Amalgam

The amalgam carrier is used to insert the mixed amalgam into the preparation in small increments (Online Fig. 26.38A). Amalgam is condensed into the retention areas first by using an appropriate condenser (Online Fig. 26.38B). Next, amalgam is condensed against the mesial and distal walls of the preparation (Online Fig. 26.38C). Finally, a sufficient bulk of amalgam is placed in the central portion to allow for carving the correct contour (Online Fig. 26.38D). As the surface of the restoration becomes more convex, condensation becomes increasingly difficult. It is important to guard against the 'landsliding' of amalgam during overpacking. A large condenser or flat-bladed instrument held against amalgam may help resist pressure applied elsewhere on the restoration (Online Fig. 26.39).

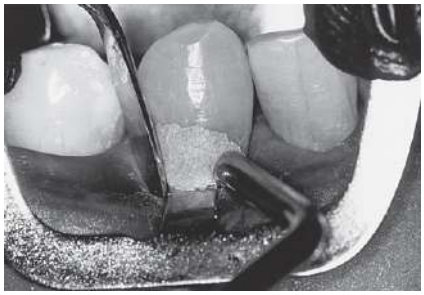
Carving may begin immediately after the insertion of amalgam (Online Fig. 26.40). All carving should be done using the side of the explorer tine or a Hollenback No. 3 carver held parallel to the margins. The side of the carving



Online Figure 26.37 Customized matrix band used to restore an area of proximal root caries. (A) Conventional Tofflemire matrix with window cut into the band to allow access for condensation. (B) Matrix in place around the tooth, allowing lingual access to preparation.

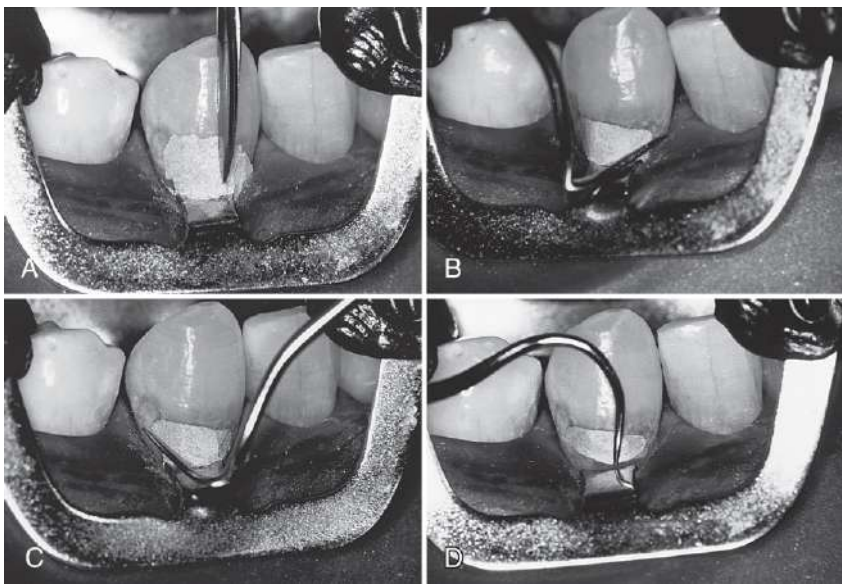


Online Figure 26.38 Inserting amalgam. (A) Place amalgam into the preparation in small increments. (B) Condense first into the retention grooves with a small condenser. (C) Condense against the mesial and distal walls. (D) Overfill and provide sufficient bulk to allow for carving.

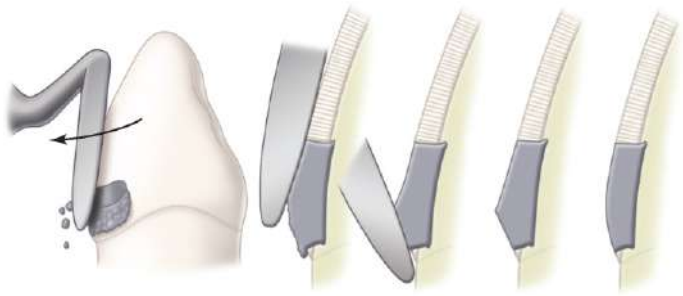


Online Figure 26.39 Use a large condenser or a flat-bladed instrument to offer resistance to condensation pressure applied elsewhere on the restoration.

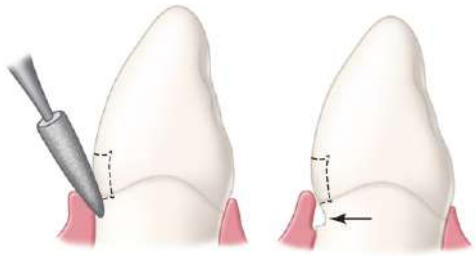
instrument always should rest on the unprepared tooth surface adjacent to the prepared cavosurface margin; this prevents overcarving. The carving procedure is begun by removing excess amalgam to expose the incisal (or occlusal) margin. Removal of excess material continues until the mesial and distal margins are exposed. Finally, material excess at the gingival margin is carved away. Carving the marginal areas should result in developing the desired convex contours in the completed restoration. Improper use of the carving instruments results in a poorly contoured restoration. Note in Online Fig. 26.41 how carving instruments are positioned to provide the desired



Online Figure 26.40 Carving and contouring the restoration. (A) Begin the carving procedure by removing any excess and locating the incisal margin. (B, C) An explorer may be used to remove the excess and locate the mesial and distal margins. (D) Remove the excess and locate the gingival margin.



Online Figure 26.41 Positioning of the carving instrument to prevent overcarving amalgam and to develop the desired gingival contours.



Online Figure 26.42 Incorrect use of a pointed stone at the gingival margin results in the removal of cementum, notching of the tooth structure gingival to the margin, or both.

contours. No amalgam excess should remain at the margins because amalgam may break away, creating a defect at the margin, or cause gingival irritation.

In some instances, it is appropriate to change facial contours because of altered soft tissue levels (e.g. cervical lesions in periodontally treated patients). Facial contours may be increased (or relocated) only enough to prevent food impaction into the gingival sulcus and to provide access for the patient to clean the area. Overcontouring must be avoided because it results in reduced stimulation and cleansing of the gingiva during mastication.

When a rubber dam and the No. 212 cervical retainer have been used for isolation, the retainer is removed using care to open the jaws of the retainer wide enough to prevent marring the surface of the restoration. The rubber dam is removed, and the area is examined carefully to ensure that no amalgam particles remain in the sulcus.

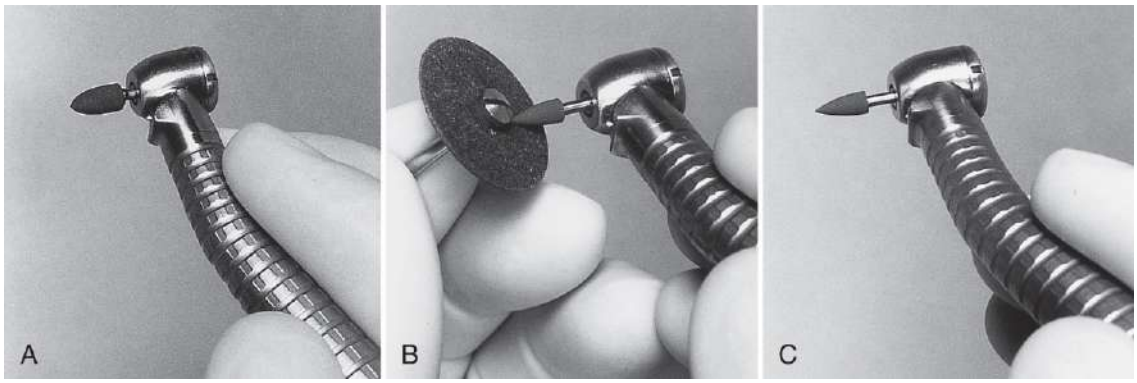
When a retraction cord is used for isolation, it may interfere with carving any excess amalgam at the gingival margin. If so, removal of the gross excess of amalgam is followed by careful removal of the cord, followed by completion of the carving along the margin.

Finishing and Polishing of the Amalgam

If carving procedures were performed correctly, no finishing of the restoration should be required. A slightly moistened cotton pellet held in cotton pliers may be used to further smooth the carved restoration. Additional finishing and polishing of amalgam restorations may be necessary, however, to correct a marginal discrepancy or to improve contour. Care is required when using stones or any rotating cutting instruments on margins positioned below the cementoenamel junction (CEJ) because of the possibility of removing cementum, notching the tooth structure gingival to the margin, or both (Online Fig. 26.42). Online Fig. 26.43 illustrates re-shaping a rubber abrasive point to allow optimal access to the gingival portion of a class V amalgam restoration.

Summary

The clinical usage of amalgam in class III and V situations is not popular as esthetic alternatives with adhesive restorations offer a more clinically acceptable choice for patients.



Online Figure 26.43 Reshaping a rubber abrasive point against a mounted carborundum disk.

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